

SUDHEER KUMAR REDDY BATTHINA

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience taking GenAI systems from concept to production, building agentic workflows, enterprise knowledge assistants, and RAG platforms that deliver measurable business outcomes rather than one-off prototypes. Skilled in designing multi-step LLM pipelines, retrieval and vector-search systems, and evaluation frameworks that keep AI outputs accurate, grounded, and reliable at scale. Strong core machine learning background spanning classification, anomaly detection, credit risk and fraud modeling, and deep learning with PyTorch and TensorFlow, applied to real-world production use cases. Experienced in MLOps practices including model deployment, CI/CD, monitoring, experimentation, and performance optimization, with a consistent focus on shipping secure, maintainable AI systems that move real business metrics.

TECHNICAL SKILLS

- **Languages & Backend:** Python, SQL, Java, C++, TypeScript, Node.js, React, Angular, Django, Spring Boot, FastAPI, REST API design, microservices, system design
- **Generative AI & Agentic Systems:** LangGraph, LangChain, LlamaIndex, OpenAI Agents SDK, CrewAI, AutoGen, multi-agent orchestration, MCP, tool and function calling, prompt engineering, Chain-of-Thought, reasoning models, guardrails, human-in-the-loop
- **RAG & LLM Evaluation:** RAG, GraphRAG, prompt optimization, rubric writing, golden response and scoring guideline development, AI output quality and accuracy assessment, vector databases (Pinecone, Weaviate, FAISS, pgvector), embeddings, LoRA/QLoRA fine-tuning, GPT, Claude, Gemini, Llama, Hugging Face Transformers, vLLM serving
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, classification, anomaly detection, reinforcement learning, NLP, feature engineering, statistical analysis, A/B testing
- **Cloud & MLOps:** AWS (Bedrock, S3, Lambda), Azure (OpenAI, AI Foundry), GCP Vertex AI, Docker, Kubernetes, CI/CD, MLflow, Ragas, DeepEval, Langfuse, OpenTelemetry, Terraform, GitOps
- **Data & Practices:** PySpark, Databricks, Power BI, Tableau, Pandas, MySQL, MongoDB, Agile/Scrum, Responsible AI

PROFESSIONAL EXPERIENCE

AI Engineer

Softtechers LLC

Jan 2025 – Present

- Architected an enterprise RAG assistant that lets scientific and clinical teams search, synthesize, and cite answers across clinical literature and internal knowledge bases, cutting evidence synthesis time by 50%+ and replacing external tools costing \$60K–\$100K annually.
- Scaled retrieval to roughly 800 QPS and cut hallucination-prone answers by 35–40% using section-aware chunking, knowledge-graph grounding, metadata filtering, hybrid retrieval, semantic reranking, and citation validation.
- Delivered an agentic workflow assistant on LangGraph, OpenAI Agents SDK, and MCP tool integrations with function calling, structured outputs, and human-in-the-loop approvals.
- Orchestrated an LLM evaluation framework with Ragas, DeepEval, and OpenTelemetry tracing, writing rubrics and golden response sets to score groundedness, accuracy, and consistency, catch hallucinations, and trace planning, retrieval, and tool-use spans, shortening release cycles by 30%.
- Cut response latency by 25–30% through FastAPI services with Redis semantic caching, async processing, and vLLM-based serving under compliance-aware controls.
- Contributed to a medical image analytics platform by building and improving PyTorch-based inference and preprocessing pipelines for image analysis workloads, reducing model inference turnaround time by 30%+.
- Fine-tuned LLMs with LoRA and QLoRA for domain adaptation and built PyTorch inference pipelines for medical imaging, reducing inference turnaround by 30%+.

Graduate Student Assistant – Data Science & AI Research

University at Buffalo

Aug 2024 – Dec 2024

- Built statistical and ML evaluation pipelines in Python, Pandas, and Scikit-learn that improved reproducibility of model comparisons and cut manual result checking by 25%.
- Developed an LLM literature search assistant with LangChain, FAISS, and Sentence Transformers, pairing semantic retrieval with agentic query planning to reduce manual search time by 35%.
- Authored preprocessing, feature analysis, and evaluation notebooks that helped faculty and student teams compare model performance consistently across experiments.

AI/ML Engineer

Wipro Limited

Dec 2020 – Mar 2024

- Drove GenAI and ML adoption across banking and financial services teams by building reusable AI components, prompt pipelines, RAG services, and evaluation logic aligned with enterprise governance and compliance requirements.
- Engineered BERT-based document classification and summarization pipelines in PyTorch for compliance review and financial transaction monitoring, decreasing manual analyst effort by 60% and improving detection accuracy across regulatory workflows.
- Led ML initiatives for credit risk scoring, customer segmentation, and fraud pattern modeling, improving risk assessment accuracy and operational efficiency by 23% across financial services client environments.
- Designed and deployed fraud detection and credit risk models using Decision Trees, Random Forests, and XGBoost to identify high-risk accounts across banking transaction datasets, reducing financial losses by 14%.
- Built CNN-based anomaly detection systems for monitoring transaction irregularities and operational risk signals, reducing manual inspection time by 42%.
- Designed repeatable ML training, validation, experiment tracking, and deployment workflows using MLflow, Airflow, Docker, Kubernetes, GitHub Actions, and Azure DevOps, reducing model release turnaround time by 28%.

PROJECTS

Enterprise Knowledge RAG Assistant – sourcebound.dev

- A production agentic RAG platform that lets enterprise users search and get source-grounded answers across internal knowledge bases, built with LangGraph multi-agent workflows, GraphRAG knowledge-graph retrieval, hybrid search, semantic reranking, and MCP tool integration, deployed on AWS with Docker and CI/CD.
- Added an evaluation-first pipeline using Ragas and DeepEval for groundedness scoring, hallucination detection, and output accuracy assessment, with Langfuse tracing and guardrails to maintain production reliability.

AI Fashion Styling and Lookbook Generator – [StyleTrip.ai](https://styletrip.ai)

- A full stack AI styling app that generates a personalized outfit lookbook using the user's own uploaded photo as the reference, giving a try-on style preview of curated looks instead of generic model images. Built with Next.js App Router, TypeScript, Tailwind CSS, and Supabase (Auth, Postgres, Storage), with a provider-routed generation pipeline supporting OpenAI, Gemini, and fal under cost guards and mock-safe defaults.
- Designed a RAG-style reference look discovery engine with deterministic match scoring and user feedback-driven re-ranking, generating frontend-rendered fashion boards exportable as PNG across multiple aspect ratios.

Location Reminder App – [RemaindAll](https://remaindall.com)

- A location reminder application that alerts users the moment they enter the radius of a place tied to a task, solving the common problem of forgetting errands like picking up groceries or dropping off a package until you're already there. Built with a React Native (Expo) mobile client, Next.js web dashboard, and Node.js (Fastify) backend, using Supabase (PostgreSQL + PostGIS) for spatial queries, Mapbox for place search, and Firebase FCM for background push notifications that fire even when the app is closed.
- Added Google Gemini 1.5 Flash function calling to let users create, update, and delete reminders conversationally, along with Siri Shortcuts and Apple CarPlay integration for hands-free reminder creation and playback while driving, plus real-time geofence trigger logging and per-user notification preferences.

EDUCATION

Master's in Data Science

State University of New York at Buffalo

Jan 2024 – May 2025